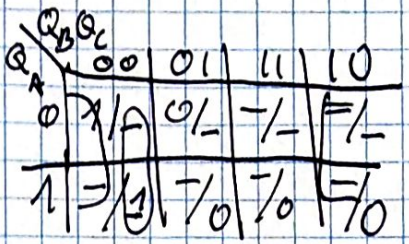
Zad 1

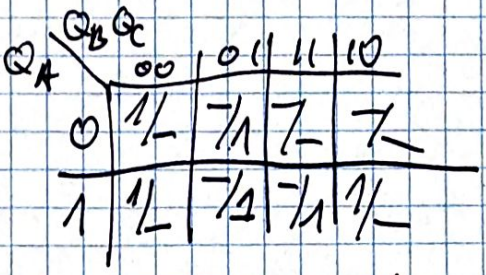
$1 \to 0 \to -1 \to -2 \to -3 \to -4 \to 1$

	Q_A	Q_B	Q_C	Q_A^+	Q_B^+	Q_C^+	J_A	K_A	J_B	K_B	J_C	K_C
0	0	0	0	1	1	1	1	-	1	-	1	-
1	0	0	1	0	1	0	0	-	0	-	-	1
2	0	1	0				-	-	-	-	-	-
3	0	1	1				-	-	-	-	-	-
-4	1	0	0	0	0	1	-	1	0	-	1	-
-3	1	0	1	1	0	0	-	0	0	-	-	1
-2	1	1	0	1	0	1	-	0	-	1	1	-
-1	1	1	1	1	1	0	-	0	-	0	-	1



$J_A = \bar{Q}_C$, $K_A = \bar{Q}_B \bar{Q}_C$





$J_B = \bar{Q}_A \bar{Q}_C$ $K_B = \bar{Q}_C$

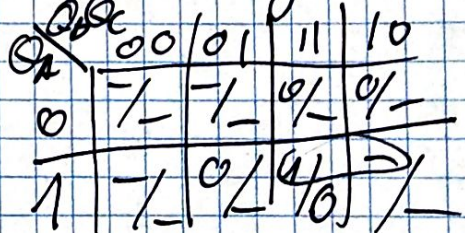
$J_C = K_C = 1$

Zad. 2.

$2 \to 3 \to 5 \to 7 \to 2$ porządek rosnący

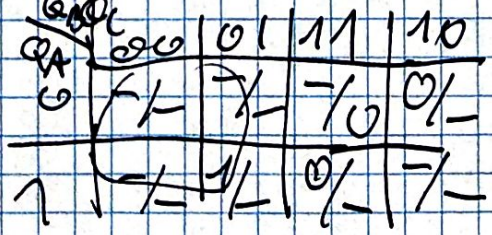
$7 \to 5 \to 3 \to 2 \to 7$ porządek malejący

	Q_A	Q_B	Q_C	Q_A^+	Q_B^+	Q_C^+	R_A	S_A	R_B	S_B	R_C	S_C
0	0	0	0	-	-	-	-	-	-	-	-	-
1	0	0	1	-	-	-	-	-	-	-	-	-
2	0	1	0	1	1	1	0	1	0	-	0	1
3	0	1	1	0	1	0	-	0	0	-	1	0
4	1	0	0	-	-	-	-	-	-	-	-	-
5	1	0	1	0	1	1	1	0	0	1	0	-
6	1	1	0	-	-	-	-	-	-	-	-	-
7	1	1	1	1	0	1	0	-	1	0	0	-



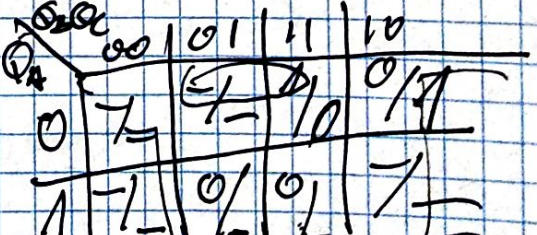
$R_B = Q_A Q_B$

$S_B = 0$



$R_A = \bar{Q}_B$

$S_A = 0$



$R_C = \bar{Q}_A Q_C$

$S_C = \bar{Q}_C$

Zad. 3.

$0 \to 1 \to 2 \to 1 \to 2 \to 3 \to 0$

$000 \to 001 \to 010 \to 101 \to 110 \to 011 \to 000$

Q_A Q_B Q_C	Q_A Q_B Q_C	$J_A K_A$	$J_B K_B$	$J_C K_C$
0 0 0	0 0 1	0 –	0 –	1 –
0 0 1	0 1 0	0 –	1 –	– 1
0 1 0	1 0 1	1 –	– 1	1 –
0 1 1	1 0 0	1 –	– 1	– 1
1 0 0	X 0 X	0 0	0 0	X X
1 0 1	1 1 0	– 0	1 –	– 1
1 1 0	0 0 1	– 1	– 1	1 –
1 1 1	0 0 0	0 X	X X	X X

Q_A \ $Q_B Q_C$	00	01	11	10
0	0/–	0/–	0/–	1/–
1	–/0	–/0	0/X	–/1

$J_A = Q_B \overline{Q_C}$

$K_A = Q_B$

Q_A \ $Q_B Q_C$	00	01	11	10
0	0/–	1/–	–/1	–/1
1	0/0	1/–	X/X	–/1

$J_B = Q_C$

$K_B = 1$

Q_A \ $Q_B Q_C$	00	01	11	10
0	1/–	–/1	–/1	1/–
1	X/X	–/1	–/–	1/–

$J_C = 1$

$K_C = 1$

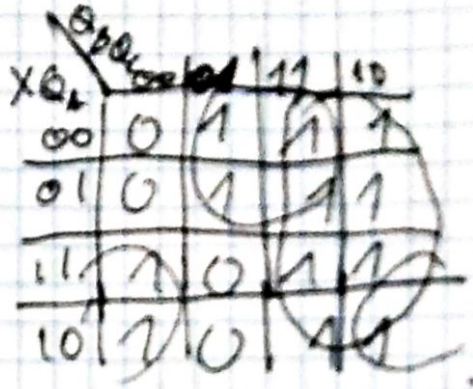
jest bezpieczny

Zad. 4.

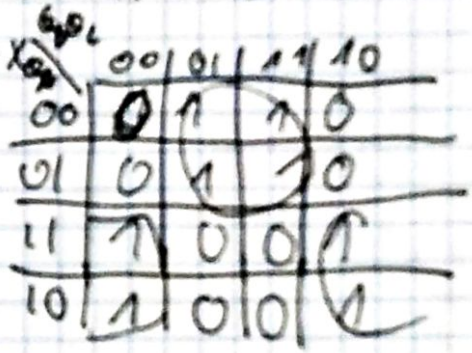
$x=0$ $0 \to 2 \to 4 \to 6 \to 0$

$x=1$ $7 \to 1 \to 3 \to 5 \to 7$

x	Q_A	Q_B	Q_C	Q_A^+	Q_B^+	Q_C^+	T_A	T_B	T_C
0	0	0	0	0	1	0	0	1	0
0	0	0	1	1	1	1	1	1	0
0	0	1	0	1	0	0	1	1	0
0	0	1	1	1	0	0	1	1	1
0	1	0	0	1	1	0	0	1	0
0	1	0	1	1	1	0	1	1	1
0	1	1	0	0	0	0	1	1	0
0	1	1	1	0	0	0	1	1	1
1	0	0	0	1	1	1	0	1	0
1	0	0	1	0	1	1	0	1	1
1	0	1	0	1	0	1	1	1	0
1	0	1	1	1	0	1	1	1	1
1	1	0	0	0	1	1	0	1	0
1	1	0	1	1	1	1	1	1	0
1	1	1	0	0	0	1	0	1	1
1	1	1	1	0	0	1	1	1	0



$T_A = Q_B + \bar{x} Q_C + x \bar{Q}_C$



$T_B = 1$

$T_C = Q_C \bar{x} + \bar{Q}_C x$